

Weiming Chen

PH.D. CANDIDATE

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Summary

I am currently a Ph.D. candidate at Southern University of Science and Technology. I am a self-driven, quick starter, passionate researcher with a curious mind who enjoys solving complex and challenging real-world computer vision problems. My research interests include machine learning, computer vision, multimodality, AIGC, text-to-image synthesis, object detection, and remote sensing.

Skills

Programming	Python (Advanced), MATLAB, C/C++, SQL
Library	PyTorch (Advanced), Tensorflow, OpenCV, Numpy, Matplotlib, Pandas, Sklearn
Front/Back-end	Hugo, Qt, Django
Design	Solidworks, AutoCAD
Languages	Chinese (Native), English (CET-6)
Others	Latex, Office

Education

Xidian (Xidian University)

Xi'an, China

B.ENG. IN MECHANICAL DESIGN MANUFACTURE AND AUTOMATION

Aug. 2015 - Jun. 2019

- Thesis: Research on Composite Sketch Recognition Algorithm Based on Deep Feature Representation. Supervisor: Prof. Xinbo Gao (IEEE Fellow)

Xidian (Xidian University)

Xi'an, China

M.SC. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Sep. 2020 - Jun. 2023

- Thesis: Research on Ship Detection Algorithm of High Resolution Optical Remote Sensing Image. Supervisor: Prof. Bing Han

SUSTech (Southern University of Science and Technology)

Shenzhen, China

PH.D. IN INTELLIGENT MANUFACTURING AND ROBOTICS

Sep. 2023 - Present

- Research Interest: Multi-modality, AIGC, Text-to-Image Synthesis. Supervisor: Prof. Zhihai He (IEEE Fellow)

Work Experience

Xi'an Air Clean Electronic Technology Co., Ltd.

Xi'an, China

CTO

Jun. 2019 - May. 2022

- Develop a remote monitoring system for mandibular postoperative rehabilitation based on the face keypoints detection algorithm.
- Develop an air quality detector.
- Responsible for the industrial design of products.

Southern University of Science and Technology

Shenzhen, China

TEACHING ASSISTANT

Sep. 2023 - Jan. 2024

- Engineering Mathematics (EE207, 2023 Fall).

Projects

Research on XXX. Project Amount: CNY 1,600,000

Xi'an, China

XXX PLAN

Oct. 2021 - Apr. 2024

- A member of the project team.
- Build a dataset of driver driving behavior based on eye movements.

Research on Key Technologies of Adaptive Multi-Domain Image Data Automatic Annotation. Project Amount: CNY 800,000

Xi'an, China

THE PROJECT OF THE KEY INDUSTRY INNOVATION CHAIN OF SHAANXI

Jan. 2022 - Dec. 2024

- The main student leader of the project.
- Responsible for the research on object incremental annotation based on domain adaptation and active learning.
- Responsible for developing a multi-domain parallel intelligent data processing platform based on artificial intelligence technology.

Cross Media Information Processing and Cognitive Curriculum Construction. Project

Amount: CNY 10,000

Xi'an, China

POSTGRADUATE CASE TEACHING CURRICULUM CONSTRUCTION PROJECT OF XIDIAN UNIVERSITY

May. 2022 - May. 2023

- The main developer of the project.
- Develop a Visual Annotation Platform based on Python and PySide6.

A Human-Machine Intention Expression and Knowledge Learning System Based on Large Models. Project Amount: CNY 20,000

Shenzhen, China

CLIMBING PLAN, UNIVERSITY STUDENTS INNOVATION AND ENTREPRENEURSHIP PROJECT IN GUANGDONG PROVINCE

Nov. 2023 - Oct. 2024

- The main leader of the project.
- Responsible for the research on multimodal human-machine interaction.
- Responsible for the research on text-to-image synthesis.

Research Project on AI-based Home Care by Zhenny Technology in the Department of Electronic and Electrical Engineering. Project Amount: CNY 4,500,000

Shenzhen, China

HORIZONTAL PROJECT

Nov. 2023 - Aug. 2024

- A member of the project team.
- Responsible for the research on real-time pose estimation and the fall-down detection algorithm.

Publications

PAPER

Weiming Chen, Zhihan Zhu, Yijia Wang, Zhihai He. Runge-Kutta Approximation and Decoupled Attention for Rectified Flow Inversion and Semantic Editing. ICLR, 2026. (Under review)

Weiming Chen, Qifan Liu, Yijia Wang, Zhihan Zhu, Siyi Liu, Yushun Tang, Zhihai He. Latent Bias Inversion: Seeking Optimal Inversion Trajectories for Real-World Images Using Diffusion Models. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025. (Under review)

Weiming Chen*, Yijia Wang*, Zhihan Zhu, Zhihai He. Generative Semantic Coding for Ultra-Low Bitrate Visual Communication and Analysis. IEEE Transactions on Image Processing, 2025. (Under review, *: Equal Contributions)

Weiming Chen*, Siqi Wu*, Yinda Chen, Dong Liu, K. C. Ho, Zhihai He. Joint Semantic and Coded Generation for Conditional Latent Coding. IEEE Transactions on Circuits and Systems for Video Technology, 2025. (Under review, *: Equal Contributions)

Siqi Wu, Yinda Chen, **Weiming Chen**, Dong Liu, K. C. Ho, Zhihai He. Learned Image Coding with Generative Reference of Conditional Latents. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2025. (Under review)

Siyi Liu*, **Weiming Chen***, Yushun Tang, Zhihai He. LatentEdit: Adaptive Latent Control for Consistent Semantic Editing[C]. PRCV, 2025. (*: Equal Contributions)

Weiming Chen, Bing Han, Zheng Yang, Xinbo Gao. MSSDet: Multi-Scale Ship-Detection Framework in Optical Remote-Sensing Images and New Benchmark[J]. Remote Sensing, 2022, 14(21): 5460. DOI: 10.3390/rs14215460. (SCI-Q2, SCI: 000883986600001, EI: 20224613127109)

Zheng Yang, Bing Han, **Weiming Chen**, Xinbo Gao. TMDiMP: Temporal Memory Guided Discriminative Tracker for UAV Object Tracking[J]. Remote Sensing. 2022, 14(24): 6351. DOI: 10.3390/rs14246351. (SCI-Q2, SCI: 000902951900-001, EI: 20225213315755)

Zheng Yang, Bing Han, **Weiming Chen**, Xinbo Gao. Learn to Encode Heterogeneous Data: A Heterogeneous Aware Network for Multi-Future Trajectory Prediction[C]. Proceedings of the International Joint Conference on Neural Networks, 2023. DOI: 10.1109/IJCNN54540.2023.10191508. (CCF-C, EI: 20233614678559)

Weiming Chen, Bing Han, Xinbo Gao. Oriented Ship Detection Based on Expansion Deformation Rotation Representation[C]. Proceedings - 2023 3rd International Conference on Frontiers of Electronics, Information and Computation Technologies, ICFEICT 2023: 621-627. DOI: 10.1109/ICFEICT59519.2023.00108. (EI: 20233914800823)

Yang Zhou, Bing Han, Xinbo Gao, Zheng Yang, **Weiming Chen**. Domain Adaptive Object Detection Based on Attention Mechanism and Cycle Domain Triplet Loss[J]. ACTA AUTOMATICA SINICA, 2024, 50(11): 2188-2203. DOI: 10.16383/j.aas.c220938.

Bing Han, Lu Gao, Xinbo Gao, **Weiming Chen**. Nuclear segmentation method for thyroid carcinoma pathologic

PATENT

Zhihai He, Siyi Liu, **Weiming Chen**, Qifan Liu, Yushun Tang, Jian Ouyang. Image inversion method, device, terminal and storage medium based on diffusion model[P]. China, Invention Patent. Application Number: 2024115602341, Application Date: 2024-11-04. (Under review)

Zhihai He, Ji yuan Jia, Yushun Tang, **Weiming Chen**, Yi Zhang, Jian Ouyang. A multi-label image classification method, device, terminal and storage medium[P]. China, Invention Patent. Application Number: 2024115602290, Application Date: 2024-11-04. Authorized Date: 2025-11-07. (Authorized)

Bing Han, Xiaoyue Huang, Zheng Yang, Xinbo Gao, **Weiming Chen**. Noise target frame correction method based on uncertainty perception and complementary information[P]. China, Invention Patent. Application Number: 2024111-666170, Application Date: 2024-08-23. Authorized Date: 2025-09-26. (Authorized)

Bing Han, **Weiming Chen**, Xinbo Gao, Zheng Yang, Xiaoyue Huang. Multi-scale target detection method based on joint recursive feature pyramid[P]. China, Invention Patent. Application Number: 2022113394400, Application Date: 2022-10-29, Authorized Date: 2025-07-04. (Authorized)

Bing Han, Zheng Yang, Xinbo Gao, **Weiming Chen**, Xiaoyue Huang. Agent trajectory prediction method based on heterogeneous data association mining and metric learning[P]. China, Invention Patent. Application Number: 2022111630697, Application Date: 2022-09-23, Authorized Date: 2025-07-01. (Authorized)

Bing Han, Zheng Yang, Xinbo Gao, **Weiming Chen**, Xiaoyue Huang. Unmanned aerial vehicle visual angle target tracking method based on sequence perception and feature enhancement[P]. China, Invention Patent. Application Number: 202211-1949471, Application Date: 2022-09-27, Authorized Date: 2025-07-01. (Authorized)

Bing Han, Lu Gao, **Weiming Chen**, Xiaoyue Huang. The invention discloses a pathological image segmentation method based on point labeling data[P]. China, Invention Patent. Application Number: 2022105696413, Application Date: 2022-05-24, Authorized Date: 2024-09-24. (Authorized)

Bing Han, **Weiming Chen**, Ping Wang, Yannan Xiong, Yang Zhou, Weixiong Zhao. Visual Image Annotation Platform V1.0[P]. China, Computer Software Copyright. Registration Number: 2023SR0432943, Registration Date: 2023-04-03. (Authorized)

Bing Han, **Weiming Chen**, Yang Zhou, Haitong Wang, Weixiong Zhao. Optical Remote Sensing Image Data Processing Platform V1.0[P]. China, Computer Software Copyright. Registration Number: 2023SR0674142, Registration Date: 2023-06-15. (Authorized)

Honors & Awards

ACADEMIC COMPETITION

2017	(National) First Prize, National College Students Advanced Mapping Technology and Product Information Modeling Innovation Competition, 3D Printing (Mechanical Group)	Lanzhou, China
2017	(National) Second Prize, National College Students Advanced Mapping Technology and Product Information Modeling Innovation Competition, Computational Modeling (Mechanical Group)	Lanzhou, China
2017	(National) Third Prize, National College Student Mathematical Modeling Challenge Competition (Undergraduate Group)	Xi'an, China
2017	(National) Third Prize, The "Teddy Cup" National College Student Data Mining Challenge Competition	Xi'an, China
2017	(Provincial) First Prize, The 5th National College Student Engineering Training Comprehensive Ability Competition Shaanxi Division Electric Control Obstacle Crossing Competition	Xi'an, China
2017	(Provincial) Second Prize, Mathematical Chinese Mathematical Modeling International Competition	Xi'an, China
2018	(School) Second Prize, Xidian University Mathematical Modeling Competition	Xi'an, China
2018	(National) Third Prize, Mathematical Contest In Modeling & Interdisciplinary Contest In Modeling	Xi'an, China
2018	(Provincial) Second Prize, The 8th National College Student Mechanical Innovation Design Competition Shaanxi Division	Xi'an, China

SCHOLARSHIPS & TITLES

2016	Second-class Scholarship , 2015-2016 academic year	<i>Xi'an, China</i>
2016	Outstanding Student Cadre , 2015-2016 academic year	<i>Xi'an, China</i>
2016	Outstanding Member of the Youth League , 2015-2016 academic year	<i>Xi'an, China</i>
2016	Outstanding Cadre of the Youth League , 2015-2016 academic year	<i>Xi'an, China</i>
2017	Special Scholarship , 2016-2017 academic year	<i>Xi'an, China</i>
2017	Outstanding Student Model , 2016-2017 academic year	<i>Xi'an, China</i>
2018	Second-class Scholarship , 2017-2018 academic year	<i>Xi'an, China</i>
2021	Second-class Scholarship , 2020-2021 academic year	<i>Xi'an, China</i>
2022	Second-class Scholarship , 2021-2022 academic year	<i>Xi'an, China</i>
2022	Outstanding Graduate Student , 2021-2022 academic year	<i>Xi'an, China</i>
2023	First-class Scholarship , 2022-2023 academic year	<i>Xi'an, China</i>
2023	Outstanding Graduate Student , 2022-2023 academic year	<i>Xi'an, China</i>

SPORTS COMPETITION

2016	(Provincial) The 3rd place , Shaanxi University Badminton Tournament, Mixed Team (Group A)	<i>Xi'an, China</i>
2017	(Provincial) The 3rd Place , Shaanxi University Badminton Tournament, Mixed Team (Group A)	<i>Xi'an, China</i>
2018	(Provincial) The 2nd Place , Shaanxi University Badminton Tournament, Mixed Team (Group A)	<i>Xi'an, China</i>
2018	(Provincial) The 6th Place , Shaanxi University Badminton Tournament, Mixed Doubles (Group A)	<i>Xi'an, China</i>
2018	(City) The 2nd Place , Decathlon Badminton Individual Challenge Competition, Men's Single	<i>Xi'an, China</i>
2019	(National) The 7th Place , Chinese College Student Sunshine Sports Badminton Competition, Men's Team	<i>Guilin, China</i>
2019	(Provincial) Champion , Shaanxi University Badminton Tournament, Men's Team (Group A)	<i>Hanzhong, China</i>
2019	(Provincial) Champion , Shaanxi University Badminton Tournament, Men's Single (Group A)	<i>Hanzhong, China</i>
2019	(City) The 3rd Place , Li-Ning 3V3 Badminton Competition, Mixed Team	<i>Xi'an, China</i>
2019	(City) The 3rd Place , Li-Ning 3V3 Badminton Competition, Mixed Doubles	<i>Xi'an, China</i>
2019	(City) The 3rd Place , Shenzhen Cup Amateur Badminton League, Mixed Team (Futian Station)	<i>Shenzhen, China</i>
2020	(Provincial) Champion , Shaanxi New Year Celebration Cup Team Tournament, Mixed Team (Group A)	<i>Xianyang, China</i>
2021	(City) Champion , YONEX "Legacy Tour" Badminton Tournament in Xi'an, Men's Doubles (Adult Group A)	<i>Xi'an, China</i>
2022	(Provincial) The 4th Place , The 17th Games of Shaanxi Province Badminton Competition, Men's Team	<i>Mizhi, China</i>
2022	(City) Champion , YONEX "Legacy Tour" Badminton Tournament in Xi'an, Mixed Doubles (Adult Group A)	<i>Xi'an, China</i>
2023	(City) Champion , Xi'an "JTHY Cup" Badminton Open, Men's Doubles (Adult Group A)	<i>Xi'an, China</i>
2025	(City) Champion , The First "ZhenXuan Cup" Badminton Team Exchange Tournament, Mixed Team	<i>Shenzhen, China</i>